

Yonatan Karp-Rudin

Staff Software Engineer, Payments · Berlin

yonvata@gmail.com

github.com/yonatankarp

linkedin.com/in/yonatankarp

Summary

Staff Software Engineer with 12 years across embedded systems, distributed backends, fintech, payments, risk, and compliance. Strongest in ambiguous domains that need service boundaries, API contracts, vendor integrations, recovery paths, and reusable tooling. Open source contributor and author on Kotlin and backend engineering.

Staff+ Focus

Defined compliance and risk service boundaries across vendor behavior, event flows, and product rules.

Drove API-first backend design with Kotlin, Java, Spring Boot, Kafka, Flink, OpenAPI, and Backstage.

Turned operational risk into recovery paths, monitoring tools, migration plans, and upstream fixes.

Experience

Staff Software Engineer - Payments

Oct 2025 - Present

Billie, Berlin, Germany

Owns payments/compliance platform reliability across service boundaries, vendor integrations, migration work, and operational tooling.

- Established Compliance Service as the backend domain for transaction monitoring, screening, and sanction checks.
- Authored technical specs and interface definitions for the ComplyAdvantage integration.

- Made failure handling explicit for Kafka DLT alerts, settlement reports, and compliance states.
- Joined the architecture team and contributes to Spring Boot 4 and Jackson 3 migration work.
- Built a Backstage cronjob monitor aggregating cross-system execution data.

Staff Software Engineer - Risk

Apr 2025 - Oct 2025

Billie, Berlin, Germany

Led Staff-level risk platform work around orchestration, fraud integrations, tokenization, and event-driven architecture.

- Co-designed and built the Risk Orchestrator with hexagonal architecture.
- Authored Buyer Risk Assessment target architecture and tokenization specs.
- Led EDA 2.0 to improve event-driven architecture practices.
- Led the Ravelin fraud integration across release planning and cross-team coordination.
- Drove OpenAPI-first design across risk services.

Senior Software Engineer - Risk

Nov 2023 - Apr 2025

Billie, Berlin, Germany

Built fraud detection and risk scoring capabilities across Flink, graph modeling, provider labelling, and Kafka infrastructure.

- Built real-time fraud velocity features per email and device with Apache Flink.
- Enabled risk scoring and provider fraud labelling for refinancing.
- Led a Neo4j graph database POC for fraud detection, then extended it with IBAN integration.
- Improved Python CI/CD pipelines and Kafka infrastructure managed through Terraform.

Backend Software Developer

Jun 2019 - Nov 2023

SumUp, Berlin, Germany

Built shared backend foundations for payments-adjacent systems, service templates, permissions, money handling, and operational reliability.

- Built a unified money library used across backend and Android.
- Designed a sales management platform with clean domain separation.
- Reduced new service setup from 1 week to 1-2 hours with a repository template.
- Created authorization and permissions management libraries.
- Migrated 6 years of historical data with minimal production impact.
- Implemented service-level objective monitoring and joined on-call for critical payment systems.

Backend Software Developer

Dec 2017 - May 2019

NICE, Ra'anana, Israel

Worked on distributed backend systems, cache reliability, and release automation.

- Built Java and Python applications on distributed systems.
- Built an automated release notes generator integrating Python 3 and Jira.
- Refactored a product cache system to improve performance and reliability.

C++/C# Junior Developer

Aug 2014 - Dec 2017

Inuitive, Ra'anana, Israel

Built production-facing embedded and tooling work for next-generation vision processing chips.

- Developed algorithms on next-generation vision processing chips in C++.
- Created calibration processes for production lines with C++, C#, Python, and Matlab.
- Helped establish production lines in South Korea and China across logistics, equipment setup, and training.

Education

Advanced Software Architecture with Residuality Theory

2024

Workshop

Advanced Software Architecture Workshop with Residuality Theory

[View credential](#)

Neo4j Fundamentals

2024

Neo4j

[View credential](#)

Software Architecture & Design of Modern Large Scale Systems

2023

Udemy

Software Architecture & Design

[View credential](#)

Design Patterns

2022

Coursera

[View credential](#)

Kotlin for Java Developers

2020

JetBrains

[View credential](#)

Reichman University

2012-2016

B.Sc. Computer Science

